Beyond the static Einstein–Euler–Heisenberg description of an extreme electric field: a formally verified validity map for discharge and trapping

Author(s) to be completed[1]

[1] *Affiliation to be completed*

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We ask what lies beyond the point where the static, quartic-truncated Einstein–Euler–Heisenberg (EEH) description of a point-like electric charge breaks down ($\varepsilon \equiv eE/m_e^2 \gtrsim 1$, $r \lesssim \lambda_C$), and whether QED can act as a "universal quantum censor" that prevents the gravitational collapse of the field. Every equation used is derived or checked with a computer algebra system (67 gates, 15 of them negative controls that must fail on altered models). We find: (i) the exact one-loop (proper-time) real part does not saturate the field; it reproduces Reissner–Nordström with a running charge, whereas the quartic truncation and a matched Born–Infeld model yield a spurious single Schwarzschild-like horizon; the dominant-energy condition is violated for $\varepsilon >1.88218$ with relative margin $\to -\alpha/3\pi$ (the one-loop trace anomaly). (ii) The condition $\theta_+ = 0 \Leftrightarrow 2GM_{\rm MS} = R$, the decomposition $M_{\rm ADM} = M_{\rm MS}(R) + U_C(R)$ and the exact cancellation of the electrostatic energy are derived inside the pipeline for time-symmetric spherical data, together with the exact one-loop correction to U_C. (iii) A Pauli-blocked Markov kinetic model keeps $0 \le f \le \frac{1}{2}$, conserves energy to 2.9×10^{-6} over at least five plasma oscillations, and gives $\omega_p = 0.186\, e\sqrt{eE_0}$ with weak damping $\gamma/\omega_p = 0.0052$. (iv) The ratio of the discharge time to the free-fall time obeys $\tau_{\rm dis}/\tau_{\rm ff} = (r_0^*/r_0)^{1/2}$ with r_0^* independent of the number of charges N, and a quantitative (N, r_0) phase map is given. Several ingredients are known results that we re-derive and verify; Sec. VI separates them from what is specific to this work, and Sec. VII states the regimes where the model is not controlled.

I. INTRODUCTION

The Euler–Heisenberg (EH) effective action [1–3] coupled to Einstein gravity has been used to study QED corrections to charged black holes in its quartic truncation [4, 5]. Its validity requires $\varepsilon \ll 1$, and for a point charge $Q = Ne$ the static field reaches $\varepsilon = 1$ at $r_\varepsilon = \sqrt{N\alpha}\,\lambda_C$, long before any horizon unless $N \gtrsim N_{\rm crit} = M_{\rm Pl}^2/m_e^2 \simeq 5.71 \times 10^{44}$. Beyond that point two qualitatively different effects matter: nonperturbative vacuum pair creation, which discharges the field on plasma time scales [6–10], and the gravitational response, which depends on whether a trapped surface can exist at all. The loss of charge by black holes through a Schwinger-type mechanism goes back to Gibbons [11], and the structure of charged thin-shell collapse to Israel, Boulware and Bekenstein [12, 13].

Aim and scope. We do not claim a new law of nature. We build a reproducible, formally gated computation that (a) quantifies where the quartic EEH extrapolation is qualitatively wrong, (b) re-derives the trapping criterion without invoking external theorems, (c) extends the pair-plasma back-reaction through several field reversals with Pauli blocking, and (d) assembles the resulting scales in one phase map. The pipeline (Appendix A) executes symbolic derivations (SymPy), adaptive numerical integrations with independent cross-checks, and negative controls; it fails if any gate fails. All numbers quoted below are read automatically from the pipeline output.

II. STATIC NONLINEAR ELECTRODYNAMICS COUPLED TO GRAVITY

A. Reduction for an arbitrary $\mathcal{L}(E)$

For the static spherical ansatz $ds^2 = -f\,dt^2 + dr^2/g + r^2 d\Omega^2$, $A = \Phi(r)\,dt$, a purely electric field $E = \sqrt{g/f}\,\Phi'$ and any Lagrangian $\mathcal{L}(X)$, $X = F_{\mu\nu}F^{\mu\nu} = -2E^2$, the stress tensor derived from the full tensor expression obeys

$$T^t{}_t = T^r{}_r = -\rho, \quad \rho = E\mathcal{L}' - \mathcal{L}, \quad T^\theta{}_\theta = \mathcal{L}, \tag{1}$$

$$r^2\mathcal{L}'(E) = Q/4\pi, \qquad f = g = 1 - \frac{2Gm(r)}{r}, \quad m' = 4\pi r^2\rho, \tag{2}$$

and $\rho' = -(2/r)(\rho + p_\perp)$ holds identically (gates S1–S4; the control C1 with $p_\perp = \mathcal{L}/2$ is detected). For Maxwell, $\rho = E^2/2$; for the quartic EH term, $\mathcal{L} = E^2/2 + 4cE^4$ with $c = \alpha^2/90\,m_e^4$.

B. One-loop proper-time Lagrangian and strong-field behaviour

For purely electric fields the spinor one-loop correction is

$$\mathcal{L}_1(x) = -\frac{1}{8\pi^2}\int_0^\infty \frac{ds}{s^3} e^{-s}\left[xs\cot(xs) - 1 + \tfrac{(xs)^2}{3}\right], \quad x = eE, \tag{3}$$

evaluated on the ray $s = \rho e^{-i\pi/4}$ because the poles of cot lie on the real axis. The real part is the effective Lagrangian; the imaginary part reproduces the Schwinger series $\mathrm{Im}\,\mathcal{L}_1 = \frac{x^2}{8\pi^3}\sum_n n^{-2} e^{-n\pi/x}$ to a relative error of

8.5×10^{-15} (gate N1), and for $\varepsilon \ll 1$ it matches $4cE^4$ with $\mathcal{L}_1/(4cE^4) - 1 \propto \varepsilon^{2.00}$ (N2; the Taylor coefficient is checked symbolically, S6).

At strong field the expansion is logarithmic, in agreement with the known renormalization-group structure [14, 15]:

$$\mathcal{L} = \frac{E^2}{2}\Big[1 - \frac{\alpha}{3\pi}\big(\ln\varepsilon + 3K\big)\Big], \quad K = -0.7640, \tag{4}$$

where the coefficient is verified symbolically (S7) and the slope $d(\mathcal{L}_1/x^2)/d\ln x = -4.22171597\times10^{-3}$ agrees with $-1/24\pi^2 = -4.22171599\times10^{-3}$ (N3). The pole of Eq. (4) sits at $\log_{10}\varepsilon_L = 561.9$, i.e. $\mu_L = \sqrt{eE} \approx 10^{277.7}$ GeV, consistent with the usual Landau scale of QED, whereas the Planck density corresponds to $\log_{10}\varepsilon_{\rm Pl} = 44.4$. This holds *at one loop*: the Ritus–Narozhny conjecture, that each loop in a strong field costs $\alpha\chi^{2/3}$ and perturbation theory fails for $\chi \gtrsim \alpha^{-3/2} \simeq 1604$ [16, 17], would invalidate all one-loop statements far below ε_L; its applicability to a pure electric field is not established here.

C. Energy conditions and the trace anomaly

WEC and NEC hold for the one-loop Lagrangian up to $\varepsilon = 10^{40}$ (the minima of ρ/E^2 and ED/E^2 are 0.465 and 0.930, gate N7), and $\mathcal{L} > 0$ (SEC) until the pole (N8). The dominant energy condition $\rho \geq |p_\perp|$ requires $ED - 2\mathcal{L} \geq 0$, and SymPy establishes the identity (gate S10)

$$ED - 2\mathcal{L} = x\mathcal{L}_1' - 2\mathcal{L}_1 = x^3\frac{d}{dx}\Big(\frac{\mathcal{L}_1}{x^2}\Big). \tag{5}$$

Hence the DEC is violated exactly where $\mathcal{L}_1/x^2$ decreases, i.e. for $\varepsilon >$1.88218 (root found with the exact integral, not the interpolation). The relative margin $(\rho - |\mathcal{L}|)/\rho$ tends to $-\alpha/3\pi = -7.743\times10^{-4}$ and keeps growing slowly with $\ln\varepsilon$ (-7.875×10^{-4} at $\varepsilon = 10^{10}$) up to the pole: it is *not* bounded. This is the one-loop trace anomaly $T^\mu{}_\mu \propto \beta F^2$ [18, 19] written in terms of ρ and $p_\perp$, not a new quantum violation. For nonlinear electrodynamics of Plebański class, causality implies the DEC [20]; the corresponding causality conditions for the full $\mathcal{L}_1(S,P)$ have not been evaluated here, so we do not draw a conclusion on (a)causality.

D. Interior profiles

Figure 1 compares the metric function and the field for $N = 10^6$ and $M = 1.5M_{\rm ext}$. The quartic extrapolation gives $E \propto r^{-2/3}$ and a finite interior mass (gate S9), and a Born–Infeld model matched to the quartic term ($b^2 = 1/32c$, S5; field bounded by $\varepsilon =$ 69.6) yields the same single horizon at $r = 2GM$, $r_h = 1.07263\times10^{-17}\,\lambda_C$. The exact one-loop result retains two horizons, $r_- = 1.48259\times10^{-18}$ and $r_+ = 9.24904\times10^{-18}$ (Reissner–Nordström: 1.36569×10^{-18} and 9.36055×10^{-18}), consistent with an effective charge $Q^2 \to \lambda Q^2$, $\lambda = E/E_{\rm Coul}(r_+) = 1.0710$: the measured shift $\delta r_+/r_+ = -1.191\times10^{-2}$ agrees with the running-charge prediction -1.230×10^{-2} (N6). The static nonlinearity therefore does not censor the horizon. These profiles are *formal solutions of the locally-constant-field model*: for $r \sim r_Q$ the field varies on a scale $\ell \simeq r_Q/2 \sim 1.8\times10^{-18}\,\lambda_C \ll \lambda_C$, outside the derivative expansion (Sec. VII).

III. TRAPPING CRITERION DERIVED INSIDE THE PIPELINE

The statement that the electrostatic energy does not help to trap a charged shell is standard in terms of the Misner–Sharp mass [12, 21, 22]; the charged Penrose inequality is a known theorem under stated hypotheses [23, 24]. To remove any dependence on citations we derive what we use (gate `P10_misner_sharp_symbolic`, 16 sub-checks).

(a) Geometry. In double-null coordinates, $ds^2 = -2e^{2\sigma}du\,dv + R^2d\Omega^2$, with $M_{\rm MS} = (R/2G)(1 - g^{ab}\partial_aR\,\partial_bR)$ and expansions $\theta_\ell = 2\partial_uR/R$, $\theta_n = 2\partial_vR/R$, SymPy proves

$$1 - \frac{2GM_{\rm MS}}{R} = -\frac{R^2e^{-2\sigma}}{2}\,\theta_\ell\theta_n. \tag{6}$$

Hence $\theta_+ = 0 \Leftrightarrow 2GM_{\rm MS} = R$ (for $\theta_- \neq 0$), and a trapped surface is $2GM_{\rm MS} > R$.

(b) Constraint. For time-symmetric data ($K_{ij} = 0$) the Hamiltonian constraint follows from computing $G_{ab}n^an^b = 2Gm'/r^2$ for a metric with arbitrary second time derivatives, which drop out; thus $m' = 4\pi r^2\rho$. The radial Maxwell field obeys Gauss' law $r^2E_{\rm orth} = $ const independently of the metric and $T_{ab}n^an^b = E_{\rm orth}^2/2$.

(c) ADM decomposition. Integrating from R to infinity,

$$M_{\rm ADM} = M_{\rm MS}(R) + U_C(R), \quad U_C(R) = \frac{Q^2}{8\pi R}, \tag{7}$$

so the exterior metric *is* Reissner–Nordström, $g = 1 - 2GM_{\rm ADM}/r + GQ^2/4\pi r^2$ (derived, not assumed).

(d) Cancellation. At $\theta_+ = 0$, $GM_{\rm ADM} = (R + r_Q^2/R)/2$ with $r_Q^2 = GQ^2/4\pi$ (the equality case of the charged Penrose inequality). With $M_{\rm MS}(R) = M_n$ the electrostatic term cancels identically: $G(M_n + U_C) - \frac{1}{2}(R + r_Q^2/R) = GM_n - R/2$. Non-charged exterior matter with $\rho \geq 0$ only adds mass, giving the inequality.

Scope. Proven: spherical time-symmetric data, $\rho \geq 0$, no charged matter outside the surface. Not proven here: the fully dynamical case (complete Misner–Sharp equations) and the persistence of trapped surfaces. The control C4 verifies that wrong factors in $M_{\rm MS}$, in the constraint ($m' = 2\pi r^2\rho$) or in U_C are detected.

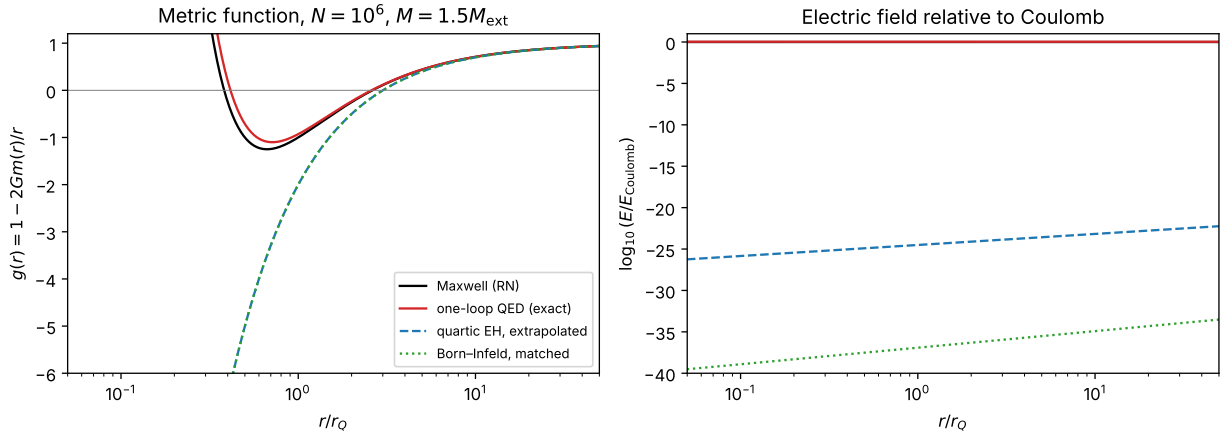


FIG. 1. Metric function $g(r) = 1 - 2Gm(r)/r$ (left) and electric field relative to Coulomb (right) for $N = 10^6$ charges and $M = 1.5M_{\rm ext}$, in the exact one-loop model, Maxwell (Reissner–Nordström), the quartic Euler–Heisenberg truncation extrapolated beyond its validity, and a Born–Infeld model matched to the quartic term.

One-loop correction to U_C. For a general $\mathcal{L}$ with $H(D) = ED - \mathcal{L}$ and $dH/dD = E$ (proved for arbitrary $\mathcal{L}$), $\rho = H(D)$, $D = a/r^2$, $a = Q/4\pi$, and

$$U(R) = 2\pi a^{3/2} \int_0^{D_R} H(D)\, D^{-5/2} dD, \quad D_R = a/R^2, \tag{8}$$

which gives $Q^2/8\pi R$ for Maxwell and $-16\pi c\, a^4/5R^5$ for the quartic term (symbolic). With the exact one-loop H and $N = 10^6$, $U_{\rm QED}/U_C - 1 = 7.047 \times 10^{-2}$ at $R = r_Q$, in agreement with a direct radial quadrature to 8.9×10^{-14} (gate P11). The slope with respect to $\ln \varepsilon$ is 7.8527×10^{-4} ($\ln \varepsilon \in [9, 23]$) and 8.1870×10^{-4} ($[18, 60]$), approaching $\alpha/3\pi = 7.7427 \times 10^{-4}$ with the $O(\alpha \ln \varepsilon)$ sub-leading logarithm.

IV. PAIR-PLASMA BACK-REACTION

A. Markov source up to the first zero

For a homogeneous field $x(t) = eE(t)$ with local pair-creation rate $\Gamma(x) = x^2 e^{-\pi/x}/4\pi^3$ (the $n = 1$ Schwinger term, which is the *mean pair number*; the vacuum-decay rate $2\,\mathrm{Im}\mathcal{L}_1$ contains the full series), conduction and polarization currents give

$$\dot{x} = -2e^2 \Big[\int_0^t dt'\, \Gamma(x') \langle v \rangle + \Gamma(x) \frac{\langle m_\perp \rangle}{x} \Big], \tag{9}$$

with $p(t, t') = \int_{t'}^t x$, $v = p/\sqrt{p^2 + m_\perp^2}$ and $m_\perp^2 = 1 + xs/\pi$ averaged over $s \sim \mathrm{Exp}(1)$. This is the Markov limit of the quantum Vlasov equation [8, 25]. The first zero of the field occurs at $\tau_{\rm dis} = \kappa/(e\sqrt{eE_0})$ with $\kappa = 13.5394$ and exponent -0.50001 (the exponent follows from scale invariance for $m \to 0$, so only κ carries information; it is specific to this model). The scheme is second order (observed order 1.98) and an independent implicit-trapezoid solver agrees ($\tau_0 = 1.41658$ vs 1.41658). As an order-of-magnitude check against a secondary report of Ref. [9], our first zero is 17.1 at $E_0 = 10E_c$ and 8.8×10^6 at $0.15E_c$ (in units of $\hbar/m_e c^2$), against ~ 10 and $\sim 10^7$ quoted there for an oscillation length; the definitions are not identical, so this is consistency, not validation.

B. Pauli-blocked kinetics through repeated field reversals

Removing the restriction to the first zero requires tracking occupation. We evolve $f(P, p_\perp, t)$ on a canonical-momentum grid with kinetic momentum $p = P + \Phi(t)$, $\dot{\Phi} = x$:

$$\partial_t f = (1 - 2f)\, S, \quad S = |x| e^{-\pi m_\perp^2/|x|}\, G_\sigma(p), \tag{10}$$

$$\dot{x} = -2e^2 \Big[\sum W f \frac{p}{\omega} + \sum W \omega \frac{(1-2f)S}{x} \Big], \tag{11}$$

with $\rho_{\rm pl} = 2\sum W f \omega$ and a Gaussian of width $\sigma = 0.25\sqrt{\varepsilon_0}$. SymPy proves that

$$\frac{d}{dt}\Big(\frac{x^2}{2e^2} + \rho_{\rm pl} \Big) = 0 \tag{12}$$

holds identically *only if* the polarization current carries the factor $(1-2f)$ (K1; the control C5 yields the residual $-4\omega S f \neq 0$ otherwise), and that the blocked equation has the exact solution $f = \frac{1}{2}(1 - e^{-2\int S}) \in [0, \frac{1}{2})$ (K2).

Results (Fig. 2). Using an adaptive DOP853 integrator with zero-crossing detection over $\varepsilon_0 \in \{30, 10^2, 10^3, 10^4\}$ we obtain at least 10 zero crossings (five full oscillations) with $\max |\Delta E|/E_0 = 2.9 \times 10^{-6} < 10^{-4}$ (K5); $0 \le f \le 0.500000$ everywhere (K4) whereas without blocking f reaches $6.53 > 1$; without the blocking factor the model reproduces the cohort model of the previous subsection (K3, $\tau_0 = 4.55758$ vs 4.55709), while

blocking lengthens the first zero to 5.5684 at $\varepsilon_0 = 100$. The plasma frequency scales as $\omega_p \propto \varepsilon_0^{0.501}$ with $\omega_p = 0.1865\, e\sqrt{eE_0}$ (K7), and the amplitude decays slowly, $\gamma/\omega_p = 0.0052$ (the amplitude falls from 0.73 to 0.62 of E_0 in nine half-cycles). Changing the tolerance, grid and transverse quadrature alters τ_0 by at most 1.5×10^{-5} (K6); varying the source width within $[0.15, 0.40]\sqrt{\varepsilon_0}$ changes it by 5.0×10^{-4}. The negative controls (no $(1-2f)$ in $j_{\rm pol}$: $|\Delta E|/E_0 = 5.4\times10^{-2}$ against 2.3×10^{-8} for the consistent model; $(1+2f)$: $f = 135 \gg \frac{1}{2}$; wrong conduction sign) are all detected.

Limitation. A Markov equation with $S \geq 0$ cannot raise f above $\frac{1}{2}$ from the vacuum, whereas the exact constant-field occupation $e^{-\pi m_\perp^2/\varepsilon}$ exceeds $\frac{1}{2}$ for $\varepsilon > 4.5$ and small $p_\perp$; for a single crossing the blocked rule gives $f = \frac{1}{2}(1-e^{-2f_0})$, which is only 0.46 of the exact value at $f_0 = 0.9$. The quantitative values of ω_p and γ are therefore properties of this model and are not compared here with non-Markovian quantum-kinetic solutions [25, 26].

V. SCALES AND PHASE MAP

For N electrons of charge Ne confined in radius r_0 (lengths in λ_C): the field reaches $\varepsilon = 1$ at $r_\varepsilon = \sqrt{N\alpha}$; the Reissner–Nordström radius is $r_Q = N\sqrt{\alpha}/M_{\rm Pl}$; the Coulomb density equals $M_{\rm Pl}^4$ at r_P (with a floor of $10\, l_P$); and the trapping radius $r_s(N)$ solves $r = 2G\,[N + \frac{3}{4}(9\pi/4)^{1/3}N^{4/3}/r]$ (rest mass plus ultrarelativistic Fermi energy, gate P4). Because of the cancellation of Sec. III, r_s is independent of the charge. The ratio of the electromagnetic to the gravitational force of the electron is $z^2 = \alpha M_{\rm Pl}^2 = 4.17 \times 10^{42}$.

Time scales. With $E_0 = Q/4\pi r_0^2$, $\varepsilon_0 = N\alpha/r_0^2$ and the result of Sec. IV, $\tau_{\rm dis} = \kappa\, r_0/(\alpha\sqrt{4\pi N})$, equal to the light-crossing time at $N^* = \kappa^2/(4\pi\alpha^2) = 2.74 \times 10^5$. For the free fall of a dust ball of mass Nm_e, $\tau_{\rm ff} = \frac{\pi}{2}\sqrt{r_0^3/2GM}$ (symbolic, P8). Both times scale as $N^{-1/2}$, so

$$\frac{\tau_{\rm dis}}{\tau_{\rm ff}} = \sqrt{\frac{r_0^*}{r_0}}, \qquad r_0^* = \frac{2\kappa^2}{\pi^3\alpha^2 M_{\rm Pl}^2} = 9.29 \times 10^{-18}\, l_P, \tag{13}$$

independent of N (gate P9). We stress that this closed form is the product of two power laws and of the electron's z^2; it only expresses that discharge is faster than free fall for every $r_0 > l_P$ (since $r_0^* \ll l_P$ the value of r_0^* itself lies outside the validity of either model). On the trapping boundary with the Fermi energy included, the discharge wins only for $N \gtrsim 1.11 \times 10^5$.

Phase map (Fig. 3). Four regions: static EFT ($\varepsilon < 1$); Schwinger discharge ($r_P < r_0 < r_\varepsilon$ and untrapped); trapping ($r_0 \leq r_s$); and trans-Planckian. The trapped region always lies at $\varepsilon \gg 1$ and above the Planck floor for $N \gtrsim 15$; a classical horizon enters the static-EFT region only for $N \gtrsim \alpha M_{\rm Pl}^4/4 = 5.94\times10^{86}$. The Coulomb density at r_Q is $\rho/M_{\rm Pl}^4 = 1/(8\pi\alpha N^2)$ (P6), sub-Planckian for $N \geq 2.3$. Table I lists the scales. The electrons in region (ii) are not bound (the Coulomb energy exceeds the Fermi energy for $N \gtrsim 6.0\times10^3$); r_s is a condition on initial data, not a formation scenario.

VI. WHAT IS KNOWN AND WHAT IS SPECIFIC TO THIS WORK

VII. REGIMES WHERE THE MODEL IS NOT CONTROLLED

(1) *Higher loops.* All strong-field statements are one-loop; Ritus–Narozhny scaling may invalidate them for $\varepsilon \gtrsim 10^3$–10^6. The sweep of the kinetic model reaches $\varepsilon_0 = 10^{10}$, 6.8 orders above that threshold. (2) *Derivative expansion and curvature.* The profiles of Sec. II descend to $r \sim 10^{-18}\lambda_C$; the curvature couplings of the one-loop action [27] are of relative order $\alpha(\lambda_C/L)^2$ and are neglected, and the local-field Lagrangian is not controlled there. (3) *Imaginary part.* For $\varepsilon > 1$ the vacuum-decay rate is $O(1)$ and the real part alone is not a stable effective action. (4) *Markov kinetics.* Homogeneous field, LCFA with $t_{\rm form}/\tau_0 \simeq e/\kappa = 0.022$, no spatial transport, $f < \frac{1}{2}$. (5) *Dynamics of trapping.* The criterion is proven for time-symmetric data; the dynamical race between discharge and collapse is not resolved spatially. (6) *Landau and Planck.* The Landau pole and the Planck density are never crossed in any calculation; the Planck floor defines region (iii) and no statement is made there.

VIII. CONCLUSIONS

Beyond the point where the quartic static EEH description fails, the exact one-loop static field does not saturate and does not censor the horizon; the electrostatic energy plays no role in trapping, which requires non-electromagnetic mass, $GM_n \geq R/2$; and pair discharge converts the field into plasma on the time $\tau_{\rm dis} = \kappa/(e\sqrt{eE_0})$ while conserving energy-momentum, so it cannot remove mass or undo a trapped surface. Within a Pauli-blocked Markov model the resulting plasma oscillates for many periods with $\omega_p \simeq 0.186\, e\sqrt{eE_0}$ and weak damping. What discharge does remove is the net charge whose repulsion exceeds gravity by $z^2 \sim 10^{42}$, and it does so faster than free fall. These conclusions are consistent with the classical picture of Misner–Sharp, Boulware and Gibbons; the added value of this work is a verified quantitative map of where each description holds, together with an open, reproducible pipeline. Non-Markovian kinetics with occupations up to unity, spatial transport and the fully dynamical trapping problem remain open.

ACKNOWLEDGMENTS

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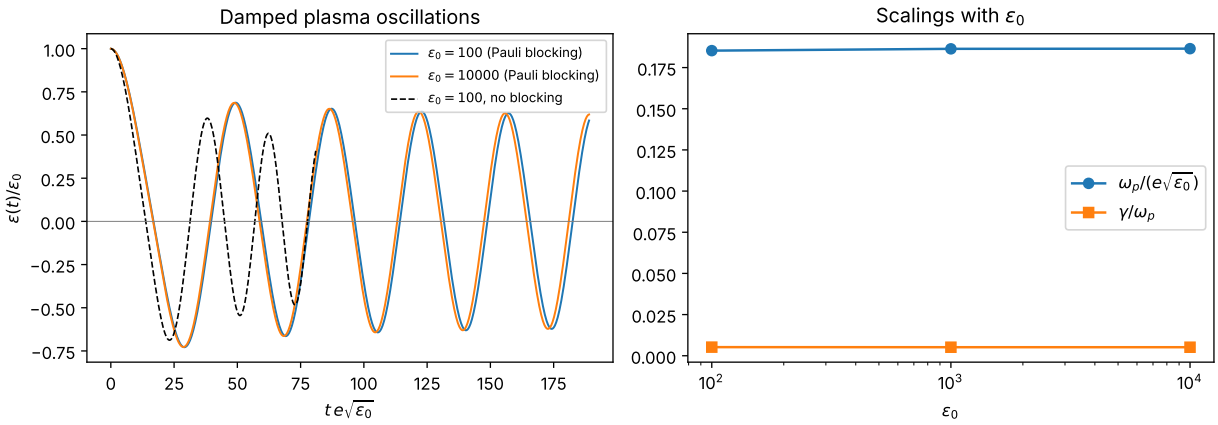


FIG. 2. Left: electric field $\varepsilon(t)/\varepsilon_0$ in units of $1/(e\sqrt{\varepsilon_0})$ for the Pauli-blocked Markov kinetic equation (solid) and without blocking (dashed), exhibiting the collapse of different ε_0 onto one curve (scale invariance). Right: $\omega_p/(e\sqrt{\varepsilon_0})$ and γ/ω_p as functions of ε_0.

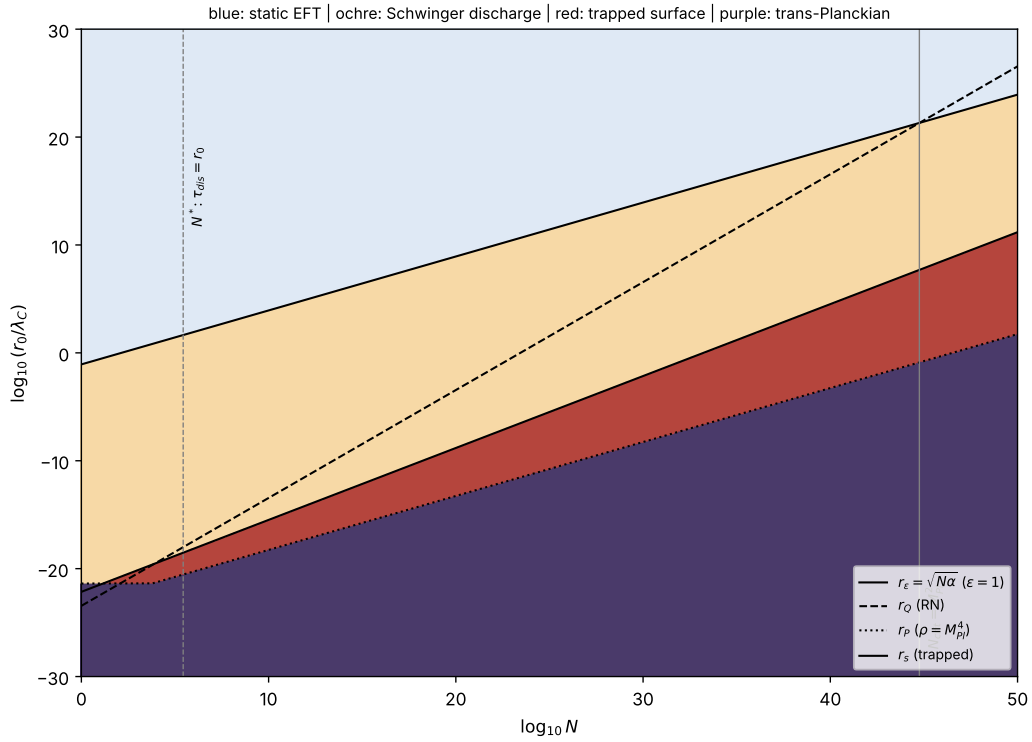


FIG. 3. Phase map in the (N, r_0) plane. Solid lines: $r_\varepsilon = \sqrt{N\alpha}$ and $r_s(N)$; dashed: r_Q; dotted: Planck-density radius. Vertical lines mark $N_{\rm crit} = M_{\rm Pl}^2/m_e^2$ and N^* ($\tau_{\rm dis} = r_0$).

thropic) operating the open pipeline described in Appendix A; the authors are responsible for the content. Bibliographic details marked in `references.bib` must be verified against the original publications before submission.

Appendix A: Verification pipeline and reproducibility

The repository (https://github.com/lopezalmeidaalvaro/experimento) contains numbered scripts executed by `run_all.sh`, which stops at the first failing gate. Symbolic gates assert simplify(LHS − RHS) = 0; numerical gates include

TABLE I. Characteristic radii (in λ_C) and ratios for selected N (computed in `06_phase_diagram.py`).

N	r_ε	r_Q	r_s	$\varepsilon(r_s)$	$\tau_{\rm dis}/r_0$
1.00×10^{0}	8.54×10^{-2}	3.58×10^{-24}	7.10×10^{-23}	1.45×10^{42}	5.23×10^{2}
1.00×10^{3}	2.70×10^{0}	3.58×10^{-21}	7.10×10^{-21}	1.45×10^{41}	1.66×10^{1}
1.00×10^{6}	8.54×10^{1}	3.58×10^{-18}	7.10×10^{-19}	1.45×10^{40}	5.23×10^{-1}
1.00×10^{12}	8.54×10^{4}	3.58×10^{-12}	7.10×10^{-15}	1.45×10^{38}	5.23×10^{-4}
1.00×10^{20}	8.54×10^{8}	3.58×10^{-4}	1.53×10^{-9}	3.12×10^{35}	5.23×10^{-8}
1.00×10^{30}	8.54×10^{13}	3.58×10^{6}	7.10×10^{-3}	1.45×10^{32}	5.23×10^{-13}
5.71×10^{44}	2.04×10^{21}	2.04×10^{21}	4.89×10^{7}	1.74×10^{27}	2.19×10^{-20}
1.00×10^{48}	8.54×10^{22}	3.58×10^{24}	7.10×10^{9}	1.45×10^{26}	5.23×10^{-22}
1.00×10^{60}	8.54×10^{28}	3.58×10^{36}	7.12×10^{17}	1.44×10^{22}	5.23×10^{-28}

TABLE II. Attribution of the results. "Known" means that the mathematical structure was published before; "here" lists what the pipeline adds.

Result	Prior work	Contribution of this work
Coulomb energy cancels in trapping	Misner–Sharp mass with charge, thin-shell energy equation, charged Penrose inequality [12, 21–24]	Derivation inside a CAS under explicit hypotheses (time-symmetric data, $\rho \geq 0$, no exterior charge); one-loop-exact $U_C(R)$
Plasma oscillations, Markov source, $\omega_p \propto e\sqrt{eE}$	[6–10, 25]	Exact energy identity with $(1-2f)$ in $j_{\rm pol}$, $f \leq \frac{1}{2}$, constants ω_p, γ for this source; convergence and negative controls
Strong-field logarithm, Landau scale	[14, 15]	Exact evaluation of the electric proper-time integral and the constant K
DEC violation	One-loop trace anomaly [18, 19]	Exact onset ε =1.88218 and identity $x^3(\mathcal{L}_1/x^2)'$
Discharge of charged black holes, $N_{\rm crit}$	Gibbons [11]	Reformulation of the scale in the (N, r_0) plane
EH black holes (quartic)	[4, 5]	Quantitative comparison with the exact one-loop interior in the coupled system, with explicit limits of validity
$\tau_{\rm dis}/\tau_{\rm ff} = (r_0^*/r_0)^{1/2}$, phase map	Elementary combination of known scalings	Closed forms with κ, verified symbolically; organization in one plane

convergence under refinement and an independent method; negative controls assert that altered signs, factors or conservation laws are detected. The totals are 67 gates, of which 15 are negative controls. Table III lists the gates by script.

TABLE III. Verification gates by script (symbolic, numerical and negative controls).

Script	gates	neg. controls
`04_nonperturbative_qed`	22	3
`05_dynamical_backreaction`	13	4
`05b_plasma_oscillations`	12	4
`06_phase_diagram`	20	4
01–03 (baseline)		re-run at each build

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